

Sanjeev Kumar Singh

📞 9973689179 | ✉️ sanjeev7363010@gmail.com | 🔗 LinkedIn | 🐙 GitHub | 📍 Patna, Bihar - 801112, India

PROFILE

Computer Science undergraduate with experience in Python, web development, and AI-based applications. Developed projects in federated learning and multi-agent systems, and eager to apply technical skills to solve real-world problems in a software development role. Eager to contribute to building scalable and efficient applications.

EDUCATION

- **National Institute of Technology Sikkim** 11/2022 – 05/2026
Bachelor of Technology in Computer Science and Engineering
CGPA: 6.84 Ravangla, India

EXPERIENCE

- **Pragament Tech Solutions Pvt. Ltd.** 06/2025 - 08/2025
Python Development Intern
 - Developed a Flask-based educational web application, integrating LLaMA 3 LLM via API to enable automated content generation, including notes, summaries, and question generation.
 - Built an Excel-like interactive frontend using HTML, CSS, and JavaScript with unit testing implemented using Mocha and Chai.
 - Implemented web scraping pipelines to collect and organize data topic- and subtopic-wise for structured access.
- **Stembotix Private Limited** 12/2024 - 02/2025
Software Development Intern
 - Learned and implemented React.js fundamentals to build responsive web application components.

PROJECTS

- **DC-SimProx: Similarity-Based Federated Learning Framework (Final Year Project)** 09/2025 – Present
Tools: Python, Machine Learning, Federated Learning, NumPy
 - Developed DC-SimProx, a dynamic clustering-based federated learning framework using similarity-aware aggregation and CountSketch signatures, reducing computational overhead by up to 95% while improving scalability.
 - Implemented robust anomaly filtering and asynchronous aggregation, enhancing model accuracy, increasing client participation by 25%, and reducing synchronization delays by 20–30% in non-IID environments.
- **Multi-Agent Essay Writer System** 10/2025 – 11/2025
Tools: Python, NLP, LLMs
 - Developed a Multi-Agent Essay Writing System using NLP & GenAI, integrating Researcher, Writer, and Editor agents to automate content generation and editing.
 - Built a Streamlit-based web application with API integration, memory handling, and PDF generation to deliver structured, editable essays based on user prompts.
- **Smart Telemedicine & Healthcare System** 🐙 GitHub 09/2025 – 10/2025
Tools: HTML, CSS, JavaScript, Python, Flask, ML Models
 - Developed a healthcare platform integrating Flask backend and JavaScript frontend.
 - Integrated prediction models for symptom analysis.

SKILLS

- **Programming Languages:** Python, SQL, JavaScript, C
- **Backend Development:** Python (Flask), REST APIs
- **Frontend Development:** HTML, CSS, JavaScript, React.js
- **Database:** MySQL, MongoDB
- **AI & LLM Technologies:** LLMs (LLaMA 3), Prompt Engineering, Multi-Agent Systems, Retrieval-Augmented Generation (RAG - concepts), LangChain (basic), NLP
- **Tools & Technologies:** Git, GitHub, VS Code
- **Core Concepts:** Object-Oriented Programming (OOP), Software Development Life Cycle (SDLC)

LANGUAGES

- English, Hindi, Bhojpuri